Note on A course in Arithmetic by J.P. Serre

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1 Chapter 1: Finite Fields and Quadratic Reciprocity

This is a list of proofs of nontrivial results in Chapter 1 of *A course in Arithmetic* by J.P. Serre.

Let K be a finite field with q elements.

Theorem 1.1

Let u be an integer ≥ 0. The sum

$$S(X^u) = \sum_{x \in K} x^u = \begin{cases} -1 & \text{if } u > 0 \text{ and } (q-1) \mid u \\ 0 & \text{otherwise} \end{cases}$$

$u = 0$ is subtle here. We define $x^0 = 1$ for all $x \in K$ (this definition is nice when we work with polynomials because we can write 1 as x^0 to maintain symmetry), so $S(X^0) = q = 0$.

The proof is easy. Use primitive roots.
The variant proof: $x \mapsto x^u$ is a group homomorphism from $K^\times$ to itself. The kernel is all x such that x is a root of $x^{\gcd(u,q-1)} - 1$. So the image has size $\frac{q-1}{\gcd(u,q-1)}$, which implies the image is just the $d = \frac{q-1}{\gcd(u,q-1)}$ roots of unity. So the sum is 0 unless $d = 1$ (sum of d-th roots of unity is 0 if $d \geq 2$ s coprime to p).

Theorem 1.2 (Chevalley-Warning Theorem)

Let $P_1, \ldots, P_r$ be polynomials in n variables over a finite field K of characteristic p. If the sum of their degrees is less than n, then the number of common solutions to the equations $P_1 = \cdots = P_r = 0$ is divisible by p.

Proof. This uses a common trick in counting roots over finite fields. Consider the polynomial

$$Q = \prod_{i=1}^{r} (1 - P_i^{q-1})$$

The number of common solutions is equal to $\sum_{x \in K^n} Q(x)$ since if $P_i(x) = 0$ for all i, then $Q(x) = 1$, otherwise $Q(x) = 0$.

Expand this. For each monomial in the expansion $x_1^{u_1} \cdots x_n^{u_n}$, since $\sum \deg(P_i)(q-1) < n(q-1)$, we have u_j is not divisible less than $q-1$ for some j. So by the previous theorem, the sum over K^n of this monomial is 0. (that's why we define $x^0 = 1$ above). So the number of common solutions is 0 mod p. $\mathfrak{Q.E.D.}$

Corollary 1.3

All quadratic forms in at least 3 variables over a finite field have nontrivial zeros.

Theorem 1.4

$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$

Proof. Let α be the primitve 8th root of unity α in $\overline{\mathbb{F}_p}$. Then, $y^2 = (\alpha+\alpha^{-1})^2 = 2$. If $p \equiv \pm 1 \mod 8$, then $y^p = y$, so 2 is a square in $\mathbb{F}_p$. If $p \equiv \pm 3 \mod 8$, then $y^p = -y$, so 2 is not a square in $\mathbb{F}_p$. $\mathfrak{Q.E.D.}$

Theorem 1.5 (Law of Quadratic Reciprocity)

For two odd primes p and q, we have

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{(p-1)(q-1)}{4}}$$

The key is to consider the Gauss sum: Let ζ be a primitive p-th root of unity in $\overline{\mathbb{F}_p}$, then define

$$y = \sum_{x \in \mathbb{F}_p} \left(\frac{x}{p}\right) \zeta^x$$

Then we have $y^2 = \left(\frac{-1}{p}\right) p = (-1)^{\frac{p-1}{2}} p$ and $y^{q-1} = \left(\frac{q}{p}\right)$. Combining these two gives the result.

2 Meeting 1 on Jan. 28 2026: Overview of the course

This is a list of nontrivial results in the first meeting of the course.

Theorem 2.1
The finite subgroup of a field's multiplicative group is cyclic.

This can be easily proven using the structure theorem of finite abelian groups and the fundamental theorem of Algebra.

Definition 2.2. For any prime p or $p = \infty$, we define the

$$\|x\|_p = \begin{cases} p^{-v_p(x)} & p \text{ is a prime} \\ \|x\| & p = \infty \end{cases}$$

Theorem 2.3
For any rational number r, $\prod_p \|r\|_p = 1$

This is trivial from the definition of v_p.

We can in general define a norm on a number field K as follows:

Definition 2.4. $\|\cdot\| : K \to \mathbb{R}_{\geq 0}$ is a norm if

1. $\|x\| = 0$ iff $x = 0$
2. $\|xy\| = \|x\| \|y\|$
3. $\|x + y\| \leq \|x\| + \|y\|$

Corollary 2.5
$\|1\| = 1$ and $\|-1\| = 1$

Definition 2.6. Norm N_1 and N_2 are equivalent if there exists a real number c such that $N_2(x) = N_1(x)^a$ for all x.

Theorem 2.7 (Ostrowski's Theorem)
Any non-trivial norm on $\mathbb{Q}$ is equivalent to either the usual absolute value $\|\cdot\|_\infty$ or to $\|\cdot\|_p$ for some prime p.

Proof. We consider two cases:

Case 1: Assumn $\|n\| \leq 1$ for all $n \in \mathbb{Z}$. If there exists n such that $\|n\| < 1$. Then there must be a prime p dividing n such that $\|p\| < 1$. We claim that p is the only prime such that $\|p\| < 1$. Let q be another prime such that $\|q\| < 1$. We can find N such that $\|p\|^N$ and $\|q\|^N$ are both less than $\frac{1}{2}$. By Bezout's lemma, we can find integers a, b such that $ap^N + bq^N = 1$. So $1 = \|1\| = \left\|ap^N + bq^N\right\| \leq \|a\| \|p\|^N + \|b\| \|q\|^N < \frac{\|a\|+\|b\|}{2} \leq 1$, which is a contradiction. So in this case, $\|\cdot\|$ is equivalent to $\|\cdot\|_p$.

The rest is sort of cumbersome. $\mathfrak{Q.E.D.}$

Example 2.8

$f(x, y, z) = 5x^2 + 7y^2 - 13z^2$. We want to find a nontrivial $f(x, y, z) = 0$ has non-trivial solutions in $\mathbb{Q}$ using Hensel's Lemma.

Consider this curve in $\mathbb{P}^3$, is there any other solution?

Use parameterization of this curve.

Theorem 2.9

If $n \geq 3$, $x^n + y^n = z^n$ has no non-trivial solutions in $\mathbb{C}[x]$.

Proof. Assume f, gh are non-constant polynomials such that $f^n + g^n = h^n$. We further assume they max $\deg(f), \deg(g), \deg(h)$ is minimal among all such triples. Therefore, $\gcd(f, g) = \gcd(g, h) = \gcd(h, f) = 1$. (otherwise, we can divide the common factor to get a smaller degree triple).

$f^n = h^n - g^n = \prod(h - \zeta^i g)$ where ζ is a primitive n-th root of unity. Notice that $\gcd(h - \zeta^i g, h - \zeta^j g) = \gcd(g, h)$ for $i \neq j$. So all $h - \zeta^i g$ are coprime to each other. Thus, $h - g = a^n$, $h - \zeta g = b^n$, $z - \zeta^2 g = c^n$ for some non-constant polynomials $a, b, c \in \mathbb{C}[x]$. We can easily find $p, q, r \in \mathbb{C}$ such that $(pa)^n + (qb)^n + (rc)^n = 0$. This contradicts the minimality of our choice of f, g, h. $\mathfrak{Q.E.D.}$

3 Chapter 2: p-adic numbers

3.1 $\mathbb{Z}_p$ and $\mathbb{Q}_p$

Let $A_n = \mathbb{Z}/p^n\mathbb{Z}$. There is a natural projection map $\phi_n : A_n \to A_{n-1}$.

Definition 3.1. We define $\mathbb{Z}_p$ as the limit (in category of rings) of

$$\cdots \xrightarrow{\phi_{n+1}} A_n \xrightarrow{\phi_n} A_{n-1} \xrightarrow{\phi_{n-1}} \cdots \xrightarrow{\phi_2} A_1$$

By the definition, any $x \in \mathbb{Z}_p$ can be represented as a seqeunce $(\ldots, x_2, x_1)$ where $x_n \in A_n$ and $\phi_n(x_n) = x_{n-1}$. Alternatively, we can represent x as an infinite sum

$$x = a_0 + a_1 p + a_2 p^2 + \cdots$$

where $a_i \in \{0, 1, \ldots, p-1\}$.

We can also define the topology on $\mathbb{Z}_p$ as the induced topology from the product topology on $\prod A_n$ where A_n is given the discrete topology.

Theorem 3.2

$\mathbb{Z}_p$ is compact.

Proof. We will need a lemma from topology.

Lemma 3.3

Closed subset of a compact space is compact.

Proof. If K is compact and $F \subseteq K$ is closed, then for any open covering of F, we can add $K \setminus F$ to get an open covering of K. So there is a finite subcovering, removing $K \setminus F$ gives a finite subcovering of F. ∎

Since $\prod A_n$ is compact by Tychonoff's theorem, it suffices to show $\mathbb{Z}_p$ is closed in $\prod A_n$. It suffices show that its complement is open. For any $y \notin \mathbb{Z}_p$, there exists some j such that $\phi_j(y_j) \neq y_{j-1}$. There exists an open set $U = \prod U_n$ where $U_n = A_n$ for $n \neq j, j-1$, $U_j = \{y_j\}$ and $U_{j-1} = \{y_{j-1}\}$. It's clear that U is an open neighborhood of y and $U \cap \mathbb{Z}_p = \emptyset$. So $\mathbb{Z}_p$ is closed. $\mathfrak{Q}.\mathfrak{E}.\mathfrak{D}.$

Theorem 3.4

$$0 \to A_m \xrightarrow{\times p^n} A_{m+n} \xrightarrow{\phi_{m+n,n}} A_n \to 0$$

is an exact seqeunce of abelian groups.

This is easy to check. The first injective map is moving the the last m digits $\times p^n$ to the left by n places and fill the last n digits with zero, the second map is just truncating the last n digits. We will use this in $\mathbb{Z}_p$.

Theorem 3.5

$$0 \to \mathbb{Z}_p \xrightarrow{\times p^n} \mathbb{Z}_p \to A_n \to 0$$

is an exact seqeunce of abelian groups.

Proof. First, we need to show $\mathbb{Z}_p \xrightarrow{\times p^n} \mathbb{Z}_p$ is injective. Notice that if $\phi_{m+n,n}(x_{m+n}) = x_n$, then $p^m x_n$ and $p^m x_{m+n}$ are the same in A_{m+n}. Therefore, we conclude that multiplying by p^n is just adding n zeros to the end, that is sending $(\dots, x_2, x_1)$ to $(\dots, p^n x_2, p^n x_1, 0, \dots, 0)$. Let j be the smallest integer such that the j-th digit of x is nonzero (not divisible by p^j), then the $j+n$-th digit of $\times p^n x$ is also nonzero since it's not divisible by p^{j+n}, so the map is injective.

The image of $\times p^n$ is just $p^n\mathbb{Z}_p$, which is exactly the kernel of the projection map to A_n since the last n digits are all zero. $\mathfrak{Q}.\mathfrak{E}.\mathfrak{D}.$

Theorem 3.6

Two propositions:

1. For an element $x \in \mathbb{Z}_p$ to be invertible, it is equivalent to say that it's not divisible by p, i.e., the first digit a_0 in its expansion is nonzero.

2. Let U be the group of units in $\mathbb{Z}_p$. Then we have the decomposition. For any $x \in \mathbb{Z}_p$, we can write

$$x = p^n u$$

where $n \geq 0$ and $u \in U$ in a unique way.

For part 1, if x_1 is invertible, a_i are corpime to p. By Bezout's lemma, there exists $a_i, b_i \in \mathbb{Z}$ such that $a_i x_i + b_i p^i = 1$. Let $a = (\dots, a_2, a_1)$, this is clearly the inverse of x and we can check it's indeed in $\mathbb{Z}_p$.

Other parts are easy.

Proposition 3.7

propositions of $\mathbb{Z}_p$ as rings:

1. As a ring, we can check that $\mathbb{Z}_p$ is an integral domain. In fact, it's a local ring with maximal ideal $p\mathbb{Z}_p$.
2. $\mathbb{Z}_p$ is discrete valuation ring. Thus, Euclidean domain, PID and UFD.

Definition 3.8. We define $v_p(x) = n$ if $x = p^n u$ where u is a unit in $\mathbb{Z}_p$. We define $v_p(0) = +\infty$.

Proposition 3.9

$v_p(xy) = v_p(x) + v_p(y)$ and $v_p(x+y) \geq \inf\{v_p(x), v_p(y)\}$ with equality if $v_p(x) \neq v_p(y)$.

This allows us to define a metric on $\mathbb{Z}_p$ as follows:

Definition 3.10. $d(x,y) = p^{-v(x-y)}$

A notbale property is that $d(x,y) = d(x-z, y-z)$ for any $x, y, z \in \mathbb{Z}_p$. So the metric is translation invariant. In particular, we have $d(x,y) = d(0, y-x)$.

Theorem 3.11

The metric space $(\mathbb{Z}_p, d)$ is homomorphic to the topological space $\mathbb{Z}_p$ defined as the subspace of $\prod A_n$ with the product topology.

Proof. It suffices to show that the identity map from $(\mathbb{Z}_p, d)$ to $\mathbb{Z}_p$ is continuous in both directions.
For any open set $U \subseteq \mathbb{Z}_p$,

$$U = \mathbb{Z}_p \cap \left(\bigcup_{i \in I} (\prod U_{n,i}) \right) = \bigcup_{i \in I} \left((\prod U_{n,i} \cap \mathbb{Z}_p) \right)$$

where $U_{n,i} = A_n$ for all but finitely many n.

What does $T_i = \prod U_{n,i} \cap \mathbb{Z}_p$ looks like? Let N be the largest integer such that $U_{N,i} \neq A_N$. Then, T_i is the set of x such that the first N digits of x are fixed and the rest can be anything possible, that is $x = (\ldots, a_N, \ldots, a_1)$ where $a_1, \ldots, a_N$ are fixed. The preimage of T_i in $(\mathbb{Z}_p, d)$ is precisely the open ball centered at x with radius p^{-N}, that is $x + p^N \mathbb{Z}_p$.

Conversely, what does any open ball in $(\mathbb{Z}_p, d)$ look like in $\mathbb{Z}_p$? First, the open ball of 0 is just ideals $p^n\mathbb{Z}_p$. For other point, if the radius of the open abll is p^{-N}, for some $N \in \mathbb{Z}$. $d(x, y) = d(0, x - y) < p^N$ means $v_p(x - y) > N$, which means the first N digits of x and y are the same. So the open balls are the cosets of $p^n\mathbb{Z}_p$. So the preimage of this open ball is the set of y such that the first N digits are fixed, which is clearly open in $\mathbb{Z}_p$. $\mathfrak{Q}.\mathfrak{E}.\mathfrak{D}.$

Proposition 3.12

propositions of $\mathbb{Z}_p$ as a metric space:

1. image of $d(0, x)$ is discrete in $\mathbb{R}$. So $\mathbb{Z}_p$ is totally disconnected.
2. $\mathbb{Z}_p$ is complete under this metric. This follows from the fact that $\mathbb{Z}_p$ is compact.
3. $\mathbb{Z}$ is dense in $\mathbb{Z}_p$. Just find an integer with the same first n digits as the element in $\mathbb{Z}_p$ you want to approximate.

Definition 3.13. We define $\mathbb{Q}_p$ as the field of fractions of $\mathbb{Z}_p$. Equivalently, it's the localization of $\mathbb{Z}_p$ at $U = 1, p, p^2, \ldots$.

Definition 3.14. For any $x \in \mathbb{Q}_p$, we can write $x = p^n u$ where $n \in \mathbb{Z}$ and $u \in U$. We define $v_p(x) = n$ and $v_p(0) = +\infty$.

Proposition 3.15

propositions about $\mathbb{Q}_p$:

1. $x \in \mathbb{Q}_p$ is in $\mathbb{Z}_p$ iff $v_p(x) \geq 0$.
2. The metric $d(x, y) = p^{-v_p(x-y)}$ makes $\mathbb{Q}_p$ a topological field.
3. $\mathbb{Z}_p$ is a clopen subset of $\mathbb{Q}_p$.
4. $\mathbb{Q}_p$ is locally compact, that is for any $x \in \mathbb{Q}_p$, there exists open set U and compact set K such that $x \in U \subseteq K$.
5. $\mathbb{Q}$ is dense in $\mathbb{Q}_p$.

For any open neighborhood of $x \in \mathbb{Q}_p$, it looks like $x + p^n\mathbb{Z}_p$ for some n (in fact clopen). Notice that translation and scaling are homeomorphisms, so any neighborhood of x is compact. Others are easy to check.

Proposition 3.16

The metric d is an ultrametric, that is it satisfies the strong triangle inequality

$$d(x, z) \leq \max\{d(x, y), d(y, z)\}$$

for any $x, y, z \in \mathbb{Q}_p$.

As a result, $d(u_m, u_n) \leq \sup_{k \geq n} d(u_k, u_{k+1})$ for any sequence $\{u_n\}$ in $\mathbb{Q}_p$ and any $m > n$. Therefore, a sequence $\{u_n\}$ is Cauchy iff $d(u_n, u_{n+1}) \to 0$ as $n \to \infty$ iff $\lim_{n\to\infty} u_{n+1} - u_n = 0$.

3.2 p-adic equations

Lemma 3.17

Let $\cdots \to D_n \to D_{n-1} \to \cdots \to D_1$ be a projective system of non-empty finite sets. Then, the projective limit $\varprojlim D_n$ is non-empty.

Proof. If each projection map is surjective, then it's clear. So our goal is to turn each map into a surjective map. For each D_n, we define $D_{n,p}$ to be the image of D_{n+p} projected to D_n. Therefore, for a fixed n, we have a decreasing sequence of non-empty finite sets

$$D_n \supseteq D_{n,1} \supseteq D_{n,2} \supseteq \cdots$$

Since D_i is finite, there exists N such that $D_{n,N} = D_{n,N+1} = \cdots$. We define $E_n = D_{n,N}$. It's clearly that E_n is non-empty and finite. And E_n is the image of E_{n+1} projected to D_n. So we have a projective system of non-empty finite sets with surjective projection maps. Therefore, the projective limit $\varprojlim E_n$ is non-empty. It's clear that $\varprojlim E_n \subseteq \varprojlim D_n$, so $\varprojlim D_n$ is non-empty. $\mathfrak{Q.E.D.}$

Definition 3.18. If $f \in \mathbb{Z}_p[x]$ is a polynomial, for $n \geq 1$, we define $f_n \in A_n[x]$ to be the reduction of f mod p^n.

Theorem 3.19

Let S be a set of polynomials in $\mathbb{Z}_p[x_1, \dots, x_n]$. The following are equivalent:

1. There exists a common root of S in $\mathbb{Z}_p^n$.
2. For all $m \geq 1$, there exists a common root of S_m (the reduction of S mod p^m) in A_m^n.

Proof. (1) $\implies$ (2) is trivial since we can reduce the common root mod p^m to get a common root in A_m^n.

For (2) $\implies$ (1), for each $m \geq 1$, let D_m be the set of common roots of S_m. So we have a project system of non-empty finite sets $\{D_m\}$ with projection maps induced by the natural projection $A_m \to A_{m-1}$. By the previous lemma, the projective limit $\varprojlim D_m$ is non-empty. Any element in the projective limit gives a common root of S in $\mathbb{Z}_p^n$. $\mathfrak{Q.E.D.}$

Definition 3.20. $x = (x_1, \ldots, x_n) \in (\mathbb{Z}_p)^n$ is called primitive if one of x_i is a unit in $\mathbb{Z}_p$. One can similarly define primitive elements in $(A_n)^n$.

Theorem 3.21

Let S be a set of homogeneous polynomials in $\mathbb{Z}_p[x_1, \ldots, x_n]$. The following are equivalent:

1. S has a non trivial common root in $\mathbb{Q}_p^n$.
2. There exists a common primitive root of S in $\mathbb{Z}_p^n$.
3. For all $m \geq 1$, there exists a common primitive root of S_m (the reduction of S mod p^m) in A_m^n.

This is easy to check.

Lemma 3.22 (Hensel's Lemma)

Let $f \in \mathbb{Z}_p[x]$ be a polynomial. Let $a \in \mathbb{Z}_p$, $n, k \in \mathbb{Z}$ such that $0 \leq 2k < n$, $f(a) \equiv 0 \pmod{p^n}$ and $v_p(f'(a)) = k$. Then, there exists a $b \in \mathbb{Z}_p$ such that $f(b) \equiv 0 \pmod{p^{n+1}}$, $v_p(f'(b)) = k$ and $b \equiv a \pmod{p^{n-k}}$.

Our familiar version is the case when $k = 0$ and $n = 1$.

Proof. Take $b = a + p^{n-k}z$ with $z \in \mathbb{Z}_p$ so that $a \equiv b \pmod{p^{n-k}}$.

$$f(b) = f(a + p^{n-k}z) = f(a) + p^{n-k}zf'(a) + p^{2n-2k}z^2t + \cdots$$

Therefore, let $f'(a) = p^k c$ and $f(a) = p^n r$, we have

$$f(b) \equiv p^n r + p^n zc \pmod{p^{n+1}}$$

Note that the second term is suppose to be $(p^{2n-2k}z^2)\frac{f''(a)}{2}$, so where is the $\frac{1}{2}$? This is a real problem if $p = 2$ The answer is that when we expand $f(a + x)$, all coefficients of x^2 should automatically be in $\mathbb{Z}_p$, so $f''(a)$ is divisible by 2. And here we denote $t = \frac{f''(a)}{2}$

Since $p \nmid c$, we can find z such that $r + zc \equiv 0 \pmod{p}$. Lastly, we can easily check that $f'(b) = f'(a+p^{n-k}z) \equiv f'(a) \pmod{p^{n-k}}$, so $v_p(f'(b)) = k$. $\mathfrak{Q}.\mathfrak{E}.\mathfrak{D}.$

Remark 3.23. Notice that we can do this iteratively to get a solution $y \in \mathbb{Z}_p$ such that $f(y) = 0$ and $y \equiv a \pmod{p^{n-k}}$. Furthermore, y is unique.

Theorem 3.24 (Generalization of Hensel's Lemma)
Let $f \in \mathbb{Z}_p[x_1, \ldots, x_m]$ be a polynomial. Let $a = (a_1, \ldots, a_n) \in (\mathbb{Z}_p)^m$, $n, k \in \mathbb{Z}$ and $1 \leq j \leq m$ such that $0 \leq 2k < n$, $f(a) \equiv 0 \pmod{p^n}$ and $v_p\left(\frac{\partial f}{\partial x_j}(a)\right) = k$.

Then there exists $y \in \mathbb{Z}_p^m$ such that $f(y) = 0$ and $y \equiv a \pmod{p^{n-k}}$.

Proof. When $m = 1$ is just Hensel's lemma.

If $m > 1$, we can plug in $x_i = a_i$ for all $i \neq j$ and consider the polynomial $g(x_j) = f(a_1, \ldots, a_{j-1}, x_j, a_{j+1}, \ldots, a_m)$. It's clear that $g(a_j) \equiv 0 \pmod{p^n}$ and $v_p(g'(a_j)) = v_p\left(\frac{\partial f}{\partial x_j}(a)\right) = k$. So by Hensel's lemma, there exists $y_j \in \mathbb{Z}_p$ such that $g(y_j) = 0$ and $y_j \equiv a_j \pmod{p^{n-k}}$. Thus, $y = (a_1, \ldots, a_{j-1}, y_j, a_{j+1}, \ldots, a_m)$ is the desired solution and $y \equiv a \pmod{p^{n-k}}$. $\mathfrak{Q}.\mathfrak{E}.\mathfrak{D}.$

Corollary 3.25

Here are three useful special cases:

1. If $a \in (\mathbb{Z}_p)^m$ such that $f(a) \equiv 0 \pmod p$ and there exists $1 \le j \le m$ such that $\frac{\partial f}{\partial x_j}(a)$ is a unit in $\mathbb{Z}_p$, then there exists $y \in (\mathbb{Z}_p)^m$ such that $f(y) = 0$ and $y \equiv a \pmod p$. This is the case when $n = 1$ and $k = 0$.

2. Suppose $p \neq 2$, let $f(x) = \sum_{i,j} a_{ij} x_i x_j$ with $a_{ij} = a_{ji} \in \mathbb{Z}_p$. If $\det(a_{ij})$ is a unit in $\mathbb{Z}_p$ and there exists a non-zero solution $r \in (\mathbb{Z}/p\mathbb{Z})^n$ such that $f(r) \equiv t \pmod p$ for some $t \in \mathbb{Z}_p$, then there exists a non-trivial solution $y \in (\mathbb{Z}_p)^n$ such that $f(y) = t$ and $y \equiv r \pmod p$.

   This is special case of (1). To prove it, we simply need to verify there exists $1 \le j \le n$ such that $\frac{\partial f}{\partial x_j}(r)$ is a unit in $\mathbb{Z}_p$. Notice that

   $$\frac{\partial f}{\partial x_i}(r) = 2 \sum_j a_{ij} r_j$$

   The vector $(\frac{\partial f}{\partial x_1}(r), \dots, \frac{\partial f}{\partial x_n}(r))$ is just $2Ar$ where $A = (a_{ij})$. Since $\det(A)$ is a unit in $\mathbb{Z}_p$, A is invertible over $\mathbb{Z}/p\mathbb{Z}$. Since r is non-trivial, Ar is also non-trivial, so at least one of its coordinates is a unit in $\mathbb{Z}_p$.

3. If $p = 2$, let $f(x) = \sum_{i,j} a_{ij} x_i x_j$ with $a_{ij} = a_{ji} \in \mathbb{Z}_2$. If $\det(a_{ij})$ is a unit in $\mathbb{Z}_2$ and there exists a non-zero solution $r \in (\mathbb{Z}/8\mathbb{Z})^n$ such that $f(r) \equiv t \pmod 8$ for some $t \in \mathbb{Z}_2$ and $\frac{\partial f}{\partial x_j}(r) \not\equiv 0 \pmod 4$, then there exists a non-trivial solution $y \in (\mathbb{Z}_2)^n$ such that $f(y) = t$ and $y \equiv r \pmod 8$. This is the case when $n = 3$ and $k = 1$. The proof is similar to (2).

3.3 multiplicative Group of $\mathbb{Q}_p$

Let $U = (\mathbb{Z}_p)^\times$ be the group of units in $\mathbb{Z}_p$ and $U_n = 1 + p^n \mathbb{Z}_p$. U_n form a decreasing sequence of subgroups of U, therefore, we have the following projective system:

$$\cdots \to \frac{U}{U_n} \to \frac{U}{U_{n-1}} \to \cdots \to \frac{U}{U_1} \to 0$$

The limit of this projective system is clearly U.

It's easy to check that $\frac{U}{U_n} \cong (\mathbb{Z}/p^n\mathbb{Z})^\times$. In particular, $\frac{U}{U_1} \cong \mathbb{F}_p^\times$.

Lemma 3.26

$\frac{U_1}{U_n}$ is finite of order p^{n-1}.

We can construct the canonical surjection from $\frac{U}{U_n} \twoheadrightarrow \frac{U}{U_1}$. The kernel is $\frac{U_1}{U_n}$. Therefore, we know that $|\frac{U_1}{U_n}| = \frac{|\frac{U}{U_n}|}{|\frac{U}{U_1}|} = \frac{p^{n-1}(p-1)}{p-1} = p^{n-1}$.

We can also construct an explicit isomorphism from $\frac{U_n}{U_{n+1}}$ $(n \geq 1)$ to $\mathbb{Z}/p\mathbb{Z}$ sending $1+p^n x$ to $x \pmod p$. $(1+p^n x)(1+p^n y) = 1+p^n(x+y)+p^{2n}xy \equiv 1+p^n(x+y) \pmod{U_{n+1}}$. So the map is indeed a homomorphism. It's clearly surjective, and the kernel is U_{n+1}. So it's an isomorphism. So $|\frac{U_n}{U_{n+1}}| = p$.

Theorem 3.27

$U \cong \mathbb{F}_p^\times \times U_1$ as abelian groups.

Proof. First, we notice that

$$0 \to U_1 \to U \to \mathbb{F}_p^\times \to 0$$

is exact. So it suffices to show this sequence splits.

First, we consider the following exact sequence:

$$0 \to \frac{U_1}{U_n} \to \frac{U}{U_n} \to \mathbb{F}_p^\times \to 0$$

This sequnece splits since $\gcd(|\mathbb{F}_p^\times|, |\frac{U_1}{U_n}|) = \gcd(p-1, p^{n-1}) = 1$. This implies there is a injective homomorphism $\mathbb{F}_p^\times \to \frac{U}{U_n}$ for all $n \geq 1$.

Therefore, by the universal property of projective limit, we have an injective homomorphism $\mathbb{F}_p^\times \to U$ splitting the original exact sequence.

Alternative proof:

Consider equaton $x^{p-1} - 1 = 0$. In $\mathbb{Z}/p\mathbb{Z}$, it has $p-1$ distinct roots. So by Hensel's lemma, it has $p-1$ distinct roots in $\mathbb{Z}_p$ (each one of them uniquely corresponds to a root in $\mathbb{Z}/p\mathbb{Z}$), giving us a injective homomorphism from $\mathbb{F}_p^\times$ to U. So we have the decomposition $U \cong \mathbb{F}_p^\times \times U_1$. $\mathfrak{Q}.\mathfrak{E}.\mathfrak{D}.$

Corollary 3.28

$\mathbb{Q}_p$ contains all $(p-1)$-th roots of unity.

Lemma 3.29

Let $x \in U_n - U_{n+1}$. Then,
if $n \geq 1$ and $p \neq 2$, $x^p \in U_{n+1} - U_{n+2}$;
if $p = 2$ and $n \geq 2$, $x^2 \in U_{n+1} - U_{n+2}$.

Proof is just to write $x = 1 + kp^n$ where $p \nmid k$ and expand x^p using binomial theorem.

Theorem 3.30

If $p \neq 2$, $(U_1, \times) \cong (\mathbb{Z}_p, +)$. If $p = 2$, $(U_2, \times) \cong (\mathbb{Z}_2, +)$ and $U_1 \cong \mathbb{Z}/2\mathbb{Z} \times (\mathbb{Z}_2, +)$.

Proof. The lemma above tells us that $\frac{U_1}{U_{n+1}}$ is a cyclic group of order p^n. So U_1 is isomorphic to the projective limit of cyclic groups of order p^n, which is $\mathbb{Z}_p$. The case when $p = 2$ is similar, we will get $U_2 \cong \mathbb{Z}_2$.

For $U_1 \cong \mathbb{Z}/2\mathbb{Z} \times (\mathbb{Z}_2, +)$, we notice the exact sequence:

$$0 \to U_2 \to U_1 \to \frac{U_1}{U_2} \to 0$$

We know that $\frac{U_1}{U_2} \cong \mathbb{Z}/2\mathbb{Z}$. So it suffices to show this sequence splits. we manually construct an injective homomorphism from $\mathbb{Z}/2\mathbb{Z}$ to U_1 sending the non-trivial element to -1.

Alternative proof with analytic approach:

It's clear that $(\mathbb{Z}_p, +)$ is isomorphic to $(p\mathbb{Z}_p, +)$ via the map $x \mapsto px$. Then, we can define the exponential map

$$\exp : p\mathbb{Z}_p \to U_1, \quad \exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

And the logarithm map

$$\log : U_1 \to p\mathbb{Z}_p, \quad \log(1 + px) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$$

It's easy to check that they are inverses of each other. Next, we need to check that they are well-defined: Notice that Therefore, we have the isomorphism $(U_1, \times) \cong (p\mathbb{Z}_p, +) \cong (\mathbb{Z}_p, +)$. $\mathfrak{Q.E.D.}$

Theorem 3.31

$(\mathbb{Q}_p^\times, \times) \cong (\mathbb{Z}, +) \times (\mathbb{Z}/(p-1)\mathbb{Z}, +) \times (\mathbb{Z}_p, +)$ if $p \neq 2$;
$(\mathbb{Q}_2^\times, \times) \cong (\mathbb{Z}, +) \times (\mathbb{Z}/2\mathbb{Z}, +) \times (\mathbb{Z}_2, +)$.
This makes multiplication into addition, which is easier to work with in many cases such as finding squares in $\mathbb{Q}_p$.

Finally, we discuss the squares in $\mathbb{Q}_p$.

Theorem 3.32

If $p \neq 2$, $x = p^n u \in \mathbb{Q}_p^\times$ is a square iff n=$v_p(x)$ is even and $u \pmod p$ is a square in $\mathbb{F}_p^\times$.

Easy to check.

Theorem 3.33

If $p = 2$, $x = 2^n u \in \mathbb{Q}_2^\times$ is a square iff $n = v_2(x)$ is even and $u \equiv 1 \pmod 8$.

In fact, u is a square if and only if $u \in U_3$ becasue U_3 is mapped to $2ZZ_2$ (square means divisible by 2 in additon group) under the isomorphism $U_1 \cong \mathbb{Z}/2\mathbb{Z} \times (\mathbb{Z}_2, +)$.

Remark 3.34. For any prime, we conclude that $(\mathbb{Q}_p^\times)^2$ is open set in $\mathbb{Q}_p^\times$. If $x = p^{2k}u$ is a square, then there exists $n = 4k$ such that the open ball $x + p^{4k}\mathbb{Z}_p$ is contained in $(\mathbb{Q}_p^\times)^2$ becasue any element in this open ball has even valuation and its unit part is congruent to u mod p, which is a square in $\mathbb{F}_p^\times$.

4 Chapter 3: Hilbert Symbol

In this section, k denotes either $\mathbb{R}$ or $\mathbb{Q}_p$.

4.1 Local Properties

Definition 4.1. Let $a, b \in k^\times$, we define

$$(a,b) = \begin{cases} 1 \text{ if } z^2 - ax^2 - by^2 = 0 \text{ has a nontrivial solution} \\ -1 \text{ otherwise} \end{cases}$$

Proposition 4.2

Let $k_b = k(\sqrt{b})$. $(a,b) = 1$ if and only if $a \in Nk_b^\times$, then group of norms of elements of $k_b^\times$

Proof is easy.

Theorem 4.3

We give a explicit form of $(a,b)_p = (a,b)$

1. If $p \neq 2$, write $a = p^n u$ and $b = p^m v$ where $n = v_p(a)$, $m = v_p(b)$ and $u, v \in U$. Then,

$$(a,b)_p = (-1)^{nm\epsilon(p)} \left(\frac{u}{p}\right)^m \left(\frac{v}{p}\right)^n$$

where $\epsilon(n) = \frac{n-1}{2} \pmod 2$.

2. If $p = 2$, write $a = 2^n u$ and $b = 2^m v$ where $n = v_2(a)$, $m = v_2(b)$ and $u, v \in U$. Then,

$$(a,b)_2 = (-1)^{\epsilon(u)\epsilon(v) + n\omega(v) + m\omega(u)}$$

where $\epsilon(x) = \frac{x-1}{2} \pmod 2$ and $\omega(x) = \frac{x^2-1}{8} \pmod 2$.

This implies that $(a,b)_p$ is bilinear and symmetric.

Remark 4.4. In the proof of this, one step is to show that if u and v are non-square units in $\mathbb{Z}_p$, then $(u,v)_p = 1$. This is equivalent to saying that the equation $z^2 - ux^2 - vy^2 = 0$ has a non-trivial solution in $\mathbb{Z}_p$. Professor Jiang provided a unique proof of this step: Consider the equation as $z^2 - ux^2 + vy^2 = 0$. If u or v are square, then we are done. Otherwise, $\left(\frac{u}{p}\right) = \left(\frac{v}{p}\right) = -1$. So $\left(\frac{uv}{p}\right) = \left(\frac{u}{p}\right)\left(\frac{v}{p}\right) = 1$. So $\left(\frac{\frac{u}{v}}{p}\right) = 1$. So can have $z = 0$

and x, y such that $\frac{u}{v}x^2 = y^2$, giving us a non-trivial solution.

Definition 4.5. We define set V to be the set of all primes and ∞. For $v \in V$, we define the local field k_v to be $\mathbb{Q}_p$ if $v = p$ is a prime and $\mathbb{R}$ if $v = \infty$. We can also define the local Hilbert symbol $(a, b)_v$ for any $a, b \in \mathbb{Q}^\times$ as the Hilbert symbol of a, b in the local field k_v.

4.2 Global Properties

Theorem 4.6

If $a, b \in \mathbb{Q}^\times$, we have $(a, b)_v = 1$ for all but finitely many $v \in V$ and

$$\prod_{v \in V} (a, b)_v = 1$$

The proof is easy.

Remark 4.7. This theorem implies the quadratic reciprocity law. Indeed, for any two primes p and q, we have $(p, q)_v = 1$ for all $v \neq p, q$. And $(p, q)_\infty = 1$. $(p, q)_p = \left(\frac{q}{p}\right)$ and $(p, q)_q = \left(\frac{p}{q}\right)$ and $(p, q)_2 = (-1)^{\epsilon(p)\epsilon(q)}$. So the product formula implies that $\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{\epsilon(p)\epsilon(q)}$, which is exactly the quadratic reciprocity law.

Theorem 4.8

let $(a_i)_{i=1}^n$ be a finite family of elements in $\mathbb{Q}^\times$ and let $\alpha_{i,v}$ be a family of numbers equal to ± 1 for $1 \leq i \leq n$ and $v \in V$. Then, the following are equivalent:

- There exists $x \in \mathbb{Q}^\times$ such that $(a_i, x)_v = \alpha_{i,v}$ for all $1 \leq i \leq n$ and $v \in V$.
- Three conditions hold:
  1. $\alpha_{i,v} = 1$ for all but finitely many $v \in V$ and $1 \leq i \leq n$.
  2. $\prod_{v \in V} \alpha_{i,v} = 1$ for all (fixed) $1 \leq i \leq n$.
  3. For any $v \in V$, there exists $x_v \in \mathbb{Q}_v^\times$ such that $(a_i, x_v)_v = \alpha_{i,v}$ for all $1 \leq i \leq n$.

The proof is easy but a bit technical. An interesting lemma is needed in the proof:

Lemma 4.9

Let S be a finite subset of V. Assume $|S| = n$, then $\mathbb{Q}^n$ is dense in $\prod_{v\in S} \mathbb{Q}_v$.

This theorem easily follows from the Chinese Remainder Theorem.

5 Chapter 4: Quadratic Forms over $\mathbb{Q}_p$ and $\mathbb{Q}$

5.1 Quadratic Forms over arbitrary fields k

Definition

We will assume the characteristic of k is not 2 in this section and all vector spaces are finite dimensional over k.

Definition 5.1. Let V be a vector space over k. A quadratic form on V is a function $Q : V \to k$ such that

1. $Q(ax) = a^2 Q(x)$ for any $a \in k$ and $x \in V$.
2. $(x, y) \mapsto Q(x + y) - Q(x) - Q(y)$ is a bilinear form.

We call pair (V, Q) a quadratic space/module over k.

Definition 5.2. $x.y = \frac{1}{2}(Q(x + y) - Q(x) - Q(y))$ is called the bilinear form associated to Q. We can also write $Q(x) = x.x$. This establishes a one-to-one correspondence between quadratic forms and symmetric bilinear forms.

Definition 5.3. The morphism between quadratic spaces (V_1, Q_1) and (V_2, Q_2) is a linear map $f : V_1 \to V_2$ such that $Q_2(f(x)) = Q_1(x)$ for all $x \in V_1$.

Corollary 5.4

$f(x).f(y) = x.y$ for any morphism f between quadratic spaces.

Next, we will introduce the matrix representation of quadratic forms. Let V be a vector space over k with basis $\{e_1, \ldots, e_n\}$. For any $x = \sum_{i=1}^n x_i e_i$, we have

$$Q(x) = (\sum_{i=1}^n x_i e_i).(\sum_{j=1}^n x_j e_j) = \sum_{i,j} a_{ij} x_i x_j$$

where $a_{ij} = e_i.e_j$. Since $e_i.e_j = e_j.e_i$, we have $a_{ij} = a_{ji}$. So we can represent Q by the symmetric matrix $M = (a_{ij})$. Furthermore, we can see that $Q(x) = x^T M x$. In general, $x.y = x^T M y$.

Notice that the matrix representation of a quadratic form depends on the choice of basis. If we change the basis by an invertible matrix P, then the new matrix representation will be $P^T M P$. So the $\det(M)$ is well-defined up to squares in k. We call $\det(M)$ the discriminant of Q and denote it by $\mathrm{disc}(Q)$.

Orthgonality

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Definition 5.5. x and y are called orthogonal if $x.y = 0$. Let $H \subseteq V$ be a subset, we define H^0 be the set of all vectors in V orthogonal to all vectors in H, and it's clearly a vector space. We call H^0 the orthogonal complement of H.

Definition 5.6. We define $\mathrm{rad}(V)$ to be V^0 when V is the ambient space.

Remark 5.7. If $U \subseteq V$, $\mathrm{rad}(U)$ is not the same as U^0 since U^0 is the set of vectors in V orthogonal to all vectors in U, while $\mathrm{rad}(U)$ is the set of vectors in U orthogonal to all vectors in U. In fact, $\mathrm{rad}(U) = U \cap U^0$.

Definition 5.8. (V, Q) is called non-degenerate if $\mathrm{rad}(V) = 0$.

Lemma 5.9

(V, Q) is non-degenerate if and only if the matrix M representing Q is invertible if and only if $\mathrm{disc}(Q) \neq 0$.

Proof. If M is not invertible, then there exists a non-zero vector x such that $Mx = 0$. So $x.y = x^T My = 0$ for any $y \in V$, so $x \in \mathrm{rad}(V)$ and (V, Q) is degenerate. Conversely, if M is invertible, Mx can be anything, so we can always find x such that $z.x = z^T Mx \neq 0$ for any non-zero z, so $\mathrm{rad}(V) = 0$ and (V, Q) is non-degenerate. $\mathfrak{Q}.\mathfrak{E}.\mathfrak{D}.$

Let U be a subsapce of (V, Q). Let U^* be the dual space of U. We can define a linear map $q_U : V \to U^*$ sending x to the linear functional $y \mapsto x.y$. We can check that $\ker(q_U) = U^0$. In particular, if $U = V$, $\ker(q_V) = \mathrm{rad}(V)$. So if (V, Q) is non-degenerate, q_V is an isomorphism from V to V^*.

Definition 5.10. Let $U_1 \cdots U_n$ be subspaces of (V, Q). We say V is the orthogonal direct sum of $U_1, \ldots, U_n$ if $V = \bigoplus_{i=1}^n U_i$ and $U_i \perp U_j$ for any $i \neq j$. We denote this by $V = U_1 \widehat{\oplus} \cdots \widehat{\oplus} U_n$.

In this case, we have $Q(x_1 + \cdots + x_n) = Q(x_1) + \cdots + Q(x_n)$ for any $x_i \in U_i$.

Theorem 5.11

Let V be a quadratic space over k and U be the complement of $\mathrm{rad}(V)$ in V, then $V = U \widehat{\oplus} \mathrm{rad}(V)$. Furthermore, U is non-degenerate and $\mathrm{rad}(V)$ is isotropic (i.e. $Q(x) = 0$ for any $x \in \mathrm{rad}(V)$).

Proof. It's clear that $V = U \widehat{\oplus} \mathrm{rad}(V)$. Now, we show that U is non-degenerate. For any $u \in \mathrm{rad}(U) \subseteq U \subseteq V$, $u.v = 0$ for all $v \in U$. In fact, $u.(v + \mathrm{rad}(V)) = 0$

for all $v \in U$, that is, $u.v = 0$ for all $v \in V$. So $u \in \mathrm{rad}(V)$. So $u = 0$, implying $\mathrm{rad}(U) = 0$ and U is non-degenerate. Lastly, for any $x \in \mathrm{rad}(V)$, $Q(x) = x.x = 0$, so $\mathrm{rad}(V)$ is isotropic. $\mathfrak{Q.E.D.}$

Proposition 5.12

Let (V, Q) be a quadratic space over k. Then,

1. All metric morphisms from (V, Q) to a quadratic space (V', Q') are injective.
2. For any subspace U of V, we have
$$U^{00} = U\,, \dim(U) + \dim(U^0) = \dim(V)\,, \mathrm{rad}(U) = \mathrm{rad}(U^0) = U \cap U^0$$
3. For any subspace U of V, U is non-degenerate if and only if U^0 is non-degenerate. In this case, we have $V = U \widehat{\oplus} U^0$.
4. If V is the orthogonal direct sum of two subspaces U and W, both U and W are non-degenerate.
5. Any non-degenerate subsapce U of V has a non-degenerate complement W such that $V = U \widehat{\oplus} W$.

Proof.

1. $f(x).f(y) = x.y$ for any morphism f between quadratic spaces. So if $f(x) = 0$, then $x.y = 0$ for all $y \in V$, so $x \in \mathrm{rad}(V)$. If (V, Q) is non-degenerate, $\mathrm{rad}(V) = 0$, so $x = 0$. So f is injective.
2. Since V is non-degenerate, we have a isomorphism $q_V : V \to V^*$. Therefore, we have surjection from V to U^* sending x to $y \mapsto x.y$. The kernel of this surjection is U^0. This gives us an exact sequence of k-vector spaces:
$$0 \to U^0 \to V \to U^* \to 0$$
Therefore, $\dim(U) + \dim(U^0) = \dim(V)$.

   Since $U \subseteq U^{00}$, and $\dim(U) = \dim(U^{00})$, we have $U = U^{00}$.

   For any $x \in \mathrm{rad}(U)$, $x \in U$ for sure. And $x.y = 0$ for any $y \in U$, so $x \in U^0$. Conversely, if $x \in U \cap U^0$, then for any $y \in U$, $x.y = 0$, so $x \in \mathrm{rad}(U)$. So $\mathrm{rad}(U) = U \cap U^0 = U^{00} \cap U^0 = \mathrm{rad}(U^0)$.
3. follows from (2).
4. If $V = U \widehat{\oplus} W$, then $W \subseteq U^0$ and $\dim(W) = \dim(V) - \dim(U) = \dim(U^0)$. So $W = U^0$ And $U^0 \cap U = \{0\}$ implies U^0 and U are non-degenerate.

5. follows from (3).

$\mathfrak{Q.E.D.}$

isotropic vectors

Definition 5.13. An element $x \in V$ is called isotropic if $Q(x) = x.x = 0$. A subspace U of V is called isotropic if $Q(x) = 0$ for any $x \in U$.

Lemma 5.14

The following are equivalent:

1. U is isotropic.
2. $U \subseteq U^0$.
3. $Q\mid_U = 0$

Proof. (1) to (2):$0 = (x+y).(x+y) = x.x + 2x.y + y.y = 2x.y$ for any $x, y \in U$. The rest is trivial. $\mathfrak{Q.E.D.}$

Definition 5.15. A quadratic module is having a basis formed by two isotropic vectors x and y such that $x.y \neq 0$ is called a hyperbolic plane.

Evidently, we can assume $x.y = 1$ by scaling y. And the matrix representation of a hyperbolic plane is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

In particular, a hyperbolic plane is non-degenerate.

Theorem 5.16

If x is an isotropic vector of a non-degenerate quadratic space (V, Q), then there exists a hyperbolic plane H containing x.

Proof. Since V is non-degenerate, there exists $y \in V$ such that $x.y = 1$. Our goal is to find z such that $z.z = 0$ and $z.x \neq 0$ (z and x are not linear dependent). We may try $z = ax + by$ and solve for a and b. At the end, we find that $z = 2y - (y.y)x$ works. $\mathfrak{Q.E.D.}$

The trick that replacing y with a linear combination of x and y to find an isotropic vector is very common in the theory of quadratic forms. We conclude as follows:

Lemma 5.17 (Linear Replacing Trick)

Given $x, y \in V$. If x is an isotropic vector ($x.x = 0$) and $x.y \neq 0$, then there exists $z = y - \frac{y.y}{2}x \in V$ such that $z.z = 0$ and $z.x = x.y \neq 0$.

Corollary 5.18

If nondegenerate quadratic space (V, Q) contains an isotropic vector, then $Q(V) = k$ that is, Q is surjective.

Proof. It suffices to show that Q is surjective on a hyperbolic plane. Let H be a hyperbolic plane with basis x, y such that $x.y = 1$. For any $a \in k$, we have $Q(x + \frac{a}{2}y) = a$. $\mathfrak{Q.E.D.}$

Orthgonal Basis

A basis $(e_1, \ldots, e_n)$ of V is called orthogonal if $e_i.e_j = 0$ for any $i \neq j$. and $e_i.e_i = a_i$ for some $a_i \in k$. In this case, the matrix representation of Q is diagonal:

$$\begin{pmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{pmatrix}$$

Theorem 5.19

Every quadratic space (V, Q) over k has an orthogonal basis.

Proof. We will prove this by induction on $\dim(V)$. If $\dim(V) = 0$, trivial. If $\dim(V) = 1$, any non-zero vector is an orthogonal basis. If $\dim(V) > 1$, we can decompose $V = \mathrm{rad}(V)\widehat{\oplus}U$ where U is non-degenerate.

Next, $\mathrm{rad}(V)$ is isotropic, so any basis of $\mathrm{rad}(V)$ is an orthogonal basis of $\mathrm{rad}(V)$. So if $\mathrm{rad}(V) \neq 0$, we can find an orthogonal basis of $\mathrm{rad}(V)$ and U by induction and get an orthogonal basis of V. If $\mathrm{rad}(V) = 0$ and V is nondegenerate, we can find x such that $x.x \neq 0$. Consider the orthogonal complement of x, H. H is non-degenerate because $H \cap kx = \{0\}$. So by proposition above, $V = kx\widehat{\oplus}H$. We can find an orthogonal basis of H by induction and get an orthogonal basis of V by adding x to the basis of H. $\mathfrak{Q.E.D.}$

We have a stronger result on orthogonal bases if V is non-degenerate.

Remark 5.20. In practice, we often want to find an orthogonal basis related (or contain) a given vector.

1. Assume (V, Q) is non-degenerate. If $x \in V$ such that $x.x \neq 0$, then we can find an orthogonal basis of V containing x. This is very easy, just consider the orthogonal complement H of x. We have $V = kx \widehat{\oplus} H$.

2. Assume (V, Q) is non-degenerate. If $x \in V$ such that $x.x = 0$, then we can find a hyperbolic plane H containing x. x can't be contained in any orthogonal basis since x is isotropic. But considering its hyperbolic plane H (which is always degenerate) is very useful.

3. Assume (V, Q) is degenerate. Then any isotropic vector x is contained in $\text{rad}(V)$, and any basis of $\text{rad}(V)$ is an orthogonal basis of $\text{rad}(V)$. So we can find an orthogonal basis of $\text{rad}(V)$ and extend it to an orthogonal basis of V.

Definition 5.21. Two orthogonal bases

$$e = (e_1, \ldots, e_n) \text{ and } f = (f_1, \ldots, f_n)$$

of V are called contiguous if they have an element in commone, that is, there exists $1 \leq i, j \leq n$ such that $e_i = f_j$.

Theorem 5.22

Let (V, Q) be a nondegenerate quadratic space of dimension ≥ 3. Then, any two orthogonal bases of V are connected by a chain of contiguous orthogonal bases.

We omit the proof of this theorem, which is a bit technical.

Witt's Theorem

Given two nondegenerate quadratic spaces (V_1, Q_1) and (V_2, Q_2). Let U be a subspace of V_1 and $s : U \to V_2$ be a morphism of quadratic spaces. We want to extend s as large as possible. That is, we want to find a morphism $s' : U' \to V_2$ where $U \subseteq U' \subseteq V_1$ and $s' |_U = s$.
Next, let's investigate the case when U is degenerate.

Lemma 5.23

If U is degenerate, then we can extend injective morphism $s : U \to V_2$ to an injective morphism $s' : U' \to V_2$ where U is a hyperplane of U'.

Proof. Since U is degenerate, $\text{rad}(U) \neq 0$. We can find an orthogonal basis of U, $(e_1, \ldots, e_r, e_{r+1}, \ldots, e_n)$ where $e_1, \ldots, e_r$ are isotropic vectors. Our goal is to add a proper vector $e_0 \notin U$ into our basis to get U'. The natural idea is to consider the hyperbolic plane H with basis e_0 and e_1. So we pick e_0 such that

$e_0.e_0 = 0$ and $e_0.e_1 = 1$.

Next, we need to define the image of e_0 under s', and we need to make sure it's compatible with quadratic forms. So let's look at what we need in order to make s' a morphism of quadratic spaces. For any $x \in U$, we can write $x = u + ce_0$ where $u \in U$ and $c \in k$. We have

$$Q_2(s'(x)) = Q_2(s'(u) + cS'(e_0)) = s(u).s(u) + cs(u)s'(e_0) + c^2 s'(e_0).s'(e_0)$$

This need to be the same as

$$Q_1(x) = u.u + 2cu.e_0 + c^2 e_0.e_0$$

Since $e_0.e_0 = 0$, we wish to have $s'(e_0).s'(e_0) = 0$. And $u.u = s(u).s(u)$ since s is a morphism of quadratic spaces. So all we need is $s(u).s'(e_0) = u.e_0$ for any $u \in U$.

This is not easy. The idea is that we can consider $u.e_0$ as a linear function $l(u) = u.e_0$ on U. $s(u).s'(e_0)$ is also a linear function $l'(s(u)) = s(u).s'(e_0)$ on $s(U)$. So wish there exists $s'(e_0)$ such that $l(u) = l'(s(u))$ for any $u \in U$. So we need to find $l' = l \circ s^{-1}$, which is a linear function on $s(U)$. Since V_2 is non-degenerate, there is a surjection from V_2 to the dual space of $s(U)$ sending v to the linear functional $s(u) \mapsto v.s(u)$. So we can find $s'(e_0)$ such that $u.e_0 = l(u) = l'(s(u)) = s(u).s'(e_0)$ for any $u \in U$.

But this $s'(e_0)$ may not satisfy $s'(e_0).s'(e_0) = 0$ but we know that $s'(e_0).s'(e_1) = e_0.e_1 = 1$. So we can apply the linear replacing trick to get a isotropic element, that is, we replace $s'(e_0)$ with $s'(e_0) - \frac{s'(e_0).s'(e_0)}{2} f(e_1)$. We're done. $\mathfrak{Q.E.D.}$

With the lemma above, we can easily extend s to a morphism defined on a non-degenerate subspace U' containing U by keeping extending s until we get a non-degenerate subspace. So we can assume U is non-degenerate. Can we extend s even more? Yes.

Theorem 5.24 (Witt's Theorem)
If (V_1, Q_1) and (V_2, Q_2) are isomorphic non-degenerate quadratic spaces, then any injective morphism $s : U \to V_2$ where U is a subspace of V_1 can be extended to an isomorphism from V_1 to V_2.

Proof. We will just assume $V_1 = V_2 = V$ and $Q_1 = Q_2 = Q$.

With the lemma above, we can assume U is non-degenerate.

Now, we show that we can extend U to a larger subsapce U'

We induct on $\dim(U)$. If $\dim(U) = 1$, then U is generated by a non-isotropic vector x. And $y = s(x)$ is also non-isotropic since $x.x = y.y$. Consider $\epsilon = \pm 1$ such that $z = x + \epsilon y$ is non-isotropic; otherwise, we would have $4x.x = 0$, which is impossible since x is non-isotropic. Now, we decompose $V = kz \widehat{\oplus} W$ and consider $\sigma \in \mathrm{Aut}(U)$ sending z to $-z$ and identity on W. Notice that $(x - \epsilon y).(x + \epsilon y) = 0$, so $x - \epsilon y \in W$. Thus,

$$\sigma(x + \epsilon y) = -x - \epsilon y \text{ and } \sigma(x - \epsilon y) = x - \epsilon y$$

Thus, $\sigma(x) = -\epsilon y$, and the automorphism $-\epsilon \circ \sigma$ of V extends s.

Now, if $\dim(U) > 1$, we can decompose $U = U_1 \widehat{\oplus} U_2$. By induction, we can extend s to an isomorphism of V, s_1 such that $s_1 |_{U_1} = s |_{U_1}$. Let $l = s_1^{-1} \circ s : U_1 \widehat{\oplus} U_2 \to V$. l acts as identity on U_1. So we have $l : U_1 \widehat{\oplus} U_2 \to U_1 \widehat{\oplus} U_2$ (morphism preserves orthogonality). So consider the $W = U_1^0 \supseteq U_2$ and $l(U_2)$. So we can extend l to an automorphism l' of W sending U_2 to $l(U_2)$. Now, let $s_2 = l' \widehat{\oplus} \mathrm{id}_{U_1} : U_1 \widehat{\oplus} W = V \to V$. We can easily verify that s_2 is an automorphism of V. Lastly, we can easily check $s_1 \circ s_2 \in \mathrm{Aut}(V)$ extends s.

$\mathfrak{Q.E.D.}$

Corollary 5.25

Two isomorphic subspace of a non-degenerate quadratic space have isomorphic orthogonal complements. That is $U_1^0 \cong U_2^0$ if $U_1 \cong U_2$.